S 1.6.2 i) A priori, one can say nothing about the hermiticity or lack thereof of $\Omega\Lambda$. If Ω and Λ commute then $\Omega\Lambda$ is hermitean since

$$(\Omega\Lambda)^\dagger = \Lambda^\dagger\Omega^\dagger = \Lambda\Omega = \Omega\Lambda .$$

If Ω and Λ anti-commute, then $\Omega\Lambda$ is anti-hermitean since

$$(\Omega\Lambda)^\dagger = \Lambda^\dagger\Omega^\dagger = \Lambda\Omega = -\Omega\Lambda$$

ii) $\Omega\Lambda + \Lambda\Omega$ is hermitean since

$$(\Omega\Lambda + \Lambda\Omega)^\dagger = \Lambda^\dagger\Omega^\dagger + \Omega^\dagger\Lambda^\dagger$$
$$= \Lambda\Omega + \Omega\Lambda = \Omega\Lambda + \Lambda\Omega$$

iii) $[\Omega, \Lambda]$ is anti-hermitean since

$$[\Omega,\Lambda]^\dagger = (\Omega\Lambda - \Lambda\Omega)^\dagger$$
$$= \Lambda^\dagger\Omega^\dagger - \Omega^\dagger\Lambda^\dagger = \Lambda\Omega - \Omega\Lambda$$
$$= -[\Omega,\Lambda]$$

iv) $i[\Omega,\Lambda]$ is hermitean since

$$(i[\Omega,\Lambda])^\dagger = (-i)[\Omega,\Lambda]^\dagger = i[\Omega,\Lambda]$$

S 1.8.3

$$\Omega = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/2 & -1/2 \\ 0 & -1/2 & -3/2 \end{pmatrix}$$

a

$$\det(\Omega - \omega \mathbb{1}) = (1-\omega)\left[\left(\tfrac{3}{2}-\omega\right)^2 - \tfrac{1}{4}\right]$$

$$= (1-\omega)(\omega^2 - 3\omega + 2)$$

$$= (1-\omega)(1-\omega)(2-\omega)$$

b.

$$(\Omega - 2\mathbb{1})|\omega = 2\rangle = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1/2 & -1/2 \\ 0 & -1/2 & -1/2 \end{pmatrix}\begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = 0$$

$\therefore \quad \omega_1 = 0 \qquad \omega_2 = -\omega_3$

Also $\quad \langle \omega = 2 | \omega = 2 \rangle = \omega_1^2 + \omega_2^2 + \omega_3^2 = 1$

$$\therefore \quad |\omega = 2\rangle = \frac{1}{\sqrt{2a^2}}\begin{pmatrix} 0 \\ a \\ -a \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

c.

$$(\Omega - \mathbb{1})|\omega = 1\rangle = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1/2 & -1/2 \\ 0 & -1/2 & 1/2 \end{pmatrix}\begin{pmatrix} \omega_1' \\ \omega_2' \\ \omega_3' \end{pmatrix} = 0$$

$$\therefore \qquad \omega_2' = \omega_3'$$

Also $\quad \langle \omega = 1 | \omega = 1 \rangle = \omega_1'^2 + \omega_2'^2 + \omega_3'^2 = 1$

$$\therefore \quad |\omega = 1\rangle = \frac{1}{\sqrt{b^2 + 2c^2}} \begin{pmatrix} b \\ c \\ c \end{pmatrix}$$

Note $\quad \langle \omega = 2 | \omega = 1 \rangle = 0$

for any value of c/b.

S1.8.5

a. $$\Omega\Omega^\dagger = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

b. $$\det(\Omega - \omega \mathbb{1}) = (\cos\theta - \omega)^2 + \sin^2\theta$$

$$= \omega^2 - 2\omega\cos\theta + 1 = 0$$

$$\omega_\pm = \cos\theta \pm \sqrt{\cos^2\theta - 1}$$

$$= \cos\theta \pm i\sin\theta = e^{\pm i\theta}$$

c. $$\begin{pmatrix} \cos\theta - e^{\pm i\theta} & \sin\theta \\ -\sin\theta & \cos\theta - e^{\pm i\theta} \end{pmatrix} |\omega = e^{\pm i\theta}\rangle$$

$$= \sin\theta \begin{pmatrix} \mp i & 1 \\ -1 & \mp i \end{pmatrix} |\omega = e^{\pm i\theta}\rangle$$

$$|\omega = e^{\pm i\theta}\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

$$\langle \omega = e^{i\theta} \mid \omega = e^{-i\theta} \rangle$$

$$= \frac{1}{2} \begin{pmatrix} 1 & +i \end{pmatrix}^* \begin{pmatrix} 1 \\ -i \end{pmatrix} = 0$$

d.

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ +i & -i \end{pmatrix}$$

$$U^\dagger \Omega U = \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}$$

$$= \frac{1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} \begin{pmatrix} e^{i\theta} & e^{-i\theta} \\ ie^{i\theta} & -ie^{-i\theta} \end{pmatrix}$$

$$= \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

S 1.8.8 $\quad M^i M^j + M^j M^i = 2\delta^{ij} I \quad (i,j = 1 \ldots 4)$

a. $\quad M^i M^i = I \quad$ (no sum on i)

since $M^{i\dagger} = M^i$, $\exists\, U^i$ such that

$$U^{i\dagger} M^i U^i = \mathrm{diag}(\omega_1^i, \omega_2^i \ldots \omega_n^i)$$

$$U^{i\dagger} M^i M^i U^i = U^{i\dagger} M^i U^i U^{i\dagger} M^i U^i$$

$$= U^{i\dagger} I U^i = I = \left(\mathrm{diag}(\omega_1^i \ldots \omega_n^i)\right)^2$$

$$\therefore (\omega_l^i)^2 = 1 \quad \therefore \omega_l^i = \pm 1 \qquad \begin{matrix} i = 1 \ldots 4 \\ l = 1 \ldots n \end{matrix}$$

since ω_l^i are real

b. $\quad \mathrm{Tr}(M^i) = \mathrm{Tr}(M^i M^j M^j) \qquad j \neq i$ no sum on j

$$= -\mathrm{Tr}(M^j M^i M^j)$$

$$= -\mathrm{Tr}(M^j M^j M^i) = -\mathrm{Tr}(M^i)$$

c. $\quad \mathrm{Tr}(M^i) = \omega_1^i + \omega_2^i + \ldots + \omega_n^i = 0$

since $\omega_l^i = \pm 1 \quad (l = 1 \ldots n)$,

n must be even.

S1.8.10

$$[\Omega, \Lambda] = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 0 & 3 \\ 0 & 0 & 0 \\ 3 & 0 & 3 \end{pmatrix} - \begin{pmatrix} 3 & 0 & 3 \\ 0 & 0 & 0 \\ 3 & 0 & 3 \end{pmatrix} = 0$$

Let us first find the eigenvalues and eigenvectors of Ω

$$\det(\Omega - \omega I) = \det \begin{pmatrix} 1-\omega & 0 & 1 \\ 0 & -\omega & 0 \\ 1 & 0 & 1-\omega \end{pmatrix}$$

$$= -\omega\left[(1-\omega)^2 - 1\right] = -\omega(\omega^2 - 2\omega)$$

$$= -\omega^2(\omega - 2)$$

$\omega_1 = 0$ double

$$(\Omega - \omega_1 I)|\omega_1\rangle = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} |\omega_1\rangle = 0$$

$$|\omega_1^1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \qquad |\omega_1^2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$\omega_2 = 2$ simple

$$(\Omega - \omega_2 I)\,|\omega_2\rangle = \begin{pmatrix} -1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & -1 \end{pmatrix} |\omega_2\rangle = 0$$

$$|\omega_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

Next consider the action of Λ on the eigenvectors of Ω

$$\Lambda\,|\omega_2\rangle = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} = 3\,|\omega_2\rangle$$

$$\Lambda\,|\omega_1^1\rangle = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \sqrt{2}\,|\omega_1^2\rangle$$

$$\Lambda\,|\omega_1^2\rangle = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \frac{1}{\sqrt{2}}$$

$$= |\omega_1^2\rangle + \sqrt{2}\,|\omega_1^1\rangle$$

$$\Lambda \begin{pmatrix} |\omega_1^1\rangle \\ |\omega_1^2\rangle \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & \sqrt{2} \\ \sqrt{2} & 1 \end{pmatrix}}_{\Lambda'} \begin{pmatrix} |\omega_1^1\rangle \\ |\omega_1^2\rangle \end{pmatrix}$$

$$\det(\Lambda' - \lambda I) = \det\begin{pmatrix} -\lambda & \sqrt{2} \\ \sqrt{2} & 1-\lambda \end{pmatrix}$$

$$= -\lambda + \lambda^2 - 2 = 0$$

$$\lambda = \frac{1}{2}\left(1 \pm \sqrt{1+8}\right) = 2 \text{ and } -1$$

$\lambda = 2$

$$\begin{pmatrix} -2 & \sqrt{2} \\ \sqrt{2} & -1 \end{pmatrix}\begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix} = 0$$

$$|1\rangle = \frac{1}{\sqrt{3}}\left(|\omega_1^1\rangle + \sqrt{2}\,|\omega_1^2\rangle\right) = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\Omega|1\rangle = 0|1\rangle \qquad \Lambda|1\rangle = 2|1\rangle$$

$\lambda = -1$

$$\begin{pmatrix} +1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}\begin{pmatrix} \sqrt{2} \\ -1 \end{pmatrix} = 0$$

$$|2\rangle = \frac{1}{\sqrt{3}}\left(\sqrt{2}\,|\omega_1^1\rangle - |\omega_1^2\rangle\right) = \frac{1}{\sqrt{6}}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$$

$$\Omega|2\rangle = 0|2\rangle \qquad \Lambda|2\rangle = (-1)|2\rangle$$

$$|3\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \Omega|3\rangle = 2|3\rangle, \quad \Lambda|3\rangle = 3|3\rangle$$

$$S = \begin{pmatrix} 1/\sqrt{3} & -1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & 2/\sqrt{6} & 0 \\ -1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{pmatrix}$$

$$S^{-1} \Omega S = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$S^{-1} \Lambda S = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$